

Maharashtra Public Service Commission Assessment

Question Paper Name :	CHROMA1C Workshop Superintendent QP2
Subject Name :	Assistant Professor Workshop Superintendent Maharashtra Engineering Teachers Service Group A
Creation Date :	2026-07-22 11:31:17
Duration :	60
Number of Questions :	100
Share Answer Key With Delivery Engine :	No
Actual Answer Key :	Yes

Assistant Professor Workshop Superintendent Maharashtra Engineering Teachers Service Group A

Group Number :	1
Group Id :	4410093004308
Group Maximum Duration :	0
Group Minimum Duration :	60
Show Attended Group? :	No
Edit Attended Group? :	No
Break time :	0

Assistant Professor Workshop Superintendent Maharashtra Engineering Teachers Service Group A

Section Id :	4410093034077
Section Number :	1
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	100

Number of Questions to be attempted :	100
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	4410093045076
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 1 Question Id : 4410092123579 Question Type : MCQ Option Shuffling : No

Which of the following statements best describes the rank of a matrix?

Options :

1. It is the total number of columns in the matrix.
2. It is the minimum number of nonzero elements in the matrix.
3. It is the maximum number of linearly independent rows or columns in the matrix.
4. It is the total number of rows in the matrix.

Question Number : 2 Question Id : 4410092123576 Question Type : MCQ Option Shuffling : No

A matrix A has two distinct eigenvalues. Which of the following situations guarantees that the eigenvectors corresponding to these eigenvalues form a linearly independent set, regardless of the matrix's diagonalizability?

Options :

1. The matrix is singular.
2. The matrix is symmetric.
3. The eigenvalues are distinct.
4. The determinant of the matrix is nonzero.

Question Number : 3 Question Id : 4410092123583 Question Type : MCQ Option Shuffling : No

A chemical engineer is optimizing the temperature of a reaction vessel to maximize product yield, modeled by the function $P(T) = -2T^2 + 40T + 300$, where T is the temperature in degrees Celsius. Which temperature should the engineer set to achieve the maximum yield, and why is this value justified as a maximum?

Options :

1. 0 degrees Celsius, since $P'(T)$ is zero there
2. 10 degrees Celsius, since $P''(T)$ is negative there, confirming a maximum
3. 15 degrees Celsius, since it is halfway between vertex points
4. 20 degrees Celsius, since it is the highest value in the domain

Question Number : 4 Question Id : 4410092123591 Question Type : MCQ Option Shuffling : No

Which theorem relates the flux of a vector field through a closed surface to the divergence of the field within the volume enclosed by the surface?

Options :

1. Green's theorem
2. Cauchy's Integral theorem
3. Gauss Divergence theorem
4. Stokes' theorem

Question Number : 5 Question Id : 4410091528301 Question Type : MCQ Option Shuffling : No

The value of $\int_C \vec{F} \cdot d\vec{r}$, if $\vec{F} = \langle x + y, x - y \rangle$ and C joins $(0,0)$ to $(1,1)$ along any smooth path, is _____.

Options :

1. 2
2. 1
3. 0
4. Path dependent

Question Number : 6 Question Id : 4410091321302 Question Type : MCQ Option Shuffling : No

The standard form of Bernoulli's differential equation is _____

Options :

1. $(dy/dx) + p(x)y = Q(x).y^n$
2. $(dy/dx) + p(x) = Q(x).y^n$
3. $(dy/dx) + Q(x).y^n = 0$
4. $(dy/dx) + p(x)y^n = Q(x)$

Question Number : 7 Question Id : 4410092123602 Question Type : MCQ Option Shuffling : No

A thin elastic membrane is stretched over a fixed wire frame in three-dimensional space. Which of the following statements best describes how the governing partial differential equation (PDE) determines the shape of the membrane?

Options :

1. The PDE determines only the boundary curve and does not influence the membrane inside the frame.
2. The membrane must always form a plane regardless of the boundary shape.
3. The PDE relates the local curvature of the membrane to the boundary conditions, determining its equilibrium surface throughout the region.
4. The PDE determines only the maximum height of the membrane and not its overall shape.

Question Number : 8 Question Id : 4410092123603 Question Type : MCQ Option Shuffling : No

Which of the following best describes the expectation (mean) of a probability distribution?

Options :

1. It is the sum of all probabilities in the distribution.
2. It is the sum of all possible values in the distribution.
3. It is the weighted average of all possible values, based on their probabilities.
4. It is the value with the highest probability in the distribution.

Question Number : 9 Question Id : 4410092123608 Question Type : MCQ Option Shuffling : No

An engineer is using the least squares method to fit a straight line to a set of experimental data points. Which step is most critical to ensure the resulting fitted line best represents the underlying trend of the data?

Options :

1. Minimizing the sum of absolute deviations between observed and predicted values
2. Selecting random points to draw the line through
3. Minimizing the sum of squared vertical deviations between observed and predicted values
4. Maximizing the sum of horizontal deviations

Question Number : 10 Question Id : 4410092123611 Question Type : MCQ Option Shuffling : No

Which of the following best describes the Newton-Raphson method for finding approximate solutions to nonlinear equations?

Options :

1. It uses only function values without derivatives to find roots.
2. It uses the function and its derivative to iteratively improve an initial guess.
3. It always converges to a solution regardless of the starting point.
4. It is applicable only for solving linear equations.

Question Number : 11 Question Id : 4410091470871 Question Type : MCQ Option Shuffling : No

For a simple cubic (SC) crystal structure, the coordination number and atomic packing factor (APF) are, respectively:

Options :

1. 8 and 0.68
2. 6 and 0.52
3. 12 and 0.74
4. 4 and 0.34

Question Number : 12 Question Id : 4410091340343 Question Type : MCQ Option Shuffling : No

Which crystal system has all axes unequal and all angles $\neq 90^\circ$, with examples like potassium dichromate ($K_2Cr_2O_7$) and copper sulfate ($CuSO_4$)?

Options :

1. Triclinic
2. Monoclinic
3. Hexagonal
4. Cubic

Question Number : 13 Question Id : 4410091470897 Question Type : MCQ Option Shuffling : No

Which of the following correctly represents the crystal system hierarchy based on symmetry (from highest to lowest)?

Options :

1. Cubic $\rightarrow$ Tetragonal $\rightarrow$ Hexagonal $\rightarrow$ Orthorhombic $\rightarrow$ Monoclinic $\rightarrow$ Triclinic
2. Triclinic $\rightarrow$ Monoclinic $\rightarrow$ Orthorhombic $\rightarrow$ Hexagonal $\rightarrow$ Tetragonal $\rightarrow$ Cubic
3. Cubic $\rightarrow$ Hexagonal $\rightarrow$ Tetragonal $\rightarrow$ Orthorhombic $\rightarrow$ Monoclinic $\rightarrow$ Triclinic
4. Hexagonal $\rightarrow$ Cubic $\rightarrow$ Orthorhombic $\rightarrow$ Tetragonal $\rightarrow$ Triclinic

Question Number : 14 Question Id : 4410091470907 Question Type : MCQ Option Shuffling : No

A plane in a cubic crystal cuts the crystallographic axes at intercepts of $2a$, $3b$ and $6c$. The Miller indices of this plane are:

Options :

1. (2 3 1)
2. (3 2 1)
3. (1 3 2)
4. (1 2 3)

Question Number : 15 Question Id : 4410091452932 Question Type : MCQ Option Shuffling : No

Under which of the following conditions, a triangular truss becomes statically indeterminate?

Options :

1. It has only three members.
2. An extra member is added.
3. It has all members rigid.
4. Loads are applied at joints.

Question Number : 16 Question Id : 4410091453018 Question Type : MCQ Option Shuffling : No

Which of the following is a key assumption in analysing spatial trusses?

Options :

1. Members carry bending and shear
2. Members carry axial forces only
3. Joints are rigid
4. Loads are applied at the midspan of members

Question Number : 17 Question Id : 441009934398 Question Type : MCQ Option Shuffling : No

Hoop stress in a thin cylinder is caused by:

Options :

1. external pressure
2. internal pressure
3. weight of fluid
4. friction

Question Number : 18 Question Id : 441009803410 Question Type : MCQ Option Shuffling : No

Which of the following parameters is inversely proportional to the buckling load according to Euler's theory?

Options :

1. Length
2. Area
3. Square of length
4. Moment of inertia

Question Number : 19 Question Id : 44100983521 Question Type : MCQ Option Shuffling : No

For a mild steel column with both ends hinged, the approximate minimum slenderness ratio for which Euler's formula will be valid is:

Options :

1. 50
2. 60
3. 70
4. 80

Question Number : 20 Question Id : 4410091469927 Question Type : MCQ Option Shuffling : No

In the conjugate beam method, the bending moment diagram of the real beam is treated as the _____ of the conjugate beam.

Options :

1. Load intensity
2. Shear force
3. Bending moment
4. Deflection

Question Number : 21 Question Id : 4410091469938 Question Type : MCQ Option Shuffling : No

Which of the following statements about the method of superposition is FALSE?

Options :

1. It can be used to combine effects of bending, shear, and axial loads.
2. It assumes linear load-deflection behaviour.
3. It is not applicable to statically indeterminate beams.
4. It simplifies analysis of complex loading.

Question Number : 22 Question Id : 441009274276 Question Type : MCQ Option Shuffling : No

As per Grashof's law:

Options :

1. the sum of shortest and longest links $\geq$ sum of other two links
2. shortest link + longest link $\leq$ sum of other two links
3. all links must be equal
4. only crank rotates

Question Number : 23 Question Id : 4410091471683 Question Type : MCQ Option Shuffling : No

In Klein's construction for a slider-crank mechanism, what is the significance of drawing a semicircle with radius equal to half the crank length?

Options :

1. It determines the angular velocity of the crank.
2. It locates the point to extend the connecting rod for the quadrilateral.
3. It represents the tangential acceleration of the piston.
4. It scales the Coriolis acceleration component.

Question Number : 24 Question Id : 4410091377340 Question Type : MCQ Option Shuffling : No

The effort of the Porter Governor is given by-

(Where m=Mass of each ball in kg

M=Mass of central load in kg

c=Percentage increase in speed

F=Frictional force in Newton)

Options :

1. $P=c[(m+M)g\pm F]$
2. $P=2c[(m+M)g\pm F]$
3. $P=3c[(m+M)g\pm F]$
4. $P=4c[(m+M)g\pm F]$

Question Number : 25 Question Id : 4410091377355 Question Type : MCQ Option Shuffling : No

The isochronous governor represents the condition of:

Options :

1. Infinite stability and infinite sensitivity.
2. Infinite stability and zero sensitivity.
3. Zero stability and infinite sensitivity.
4. Finite stability and finite sensitivity.

Question Number : 26 Question Id : 4410091216195 Question Type : MCQ Option Shuffling : No

According to Saint Venant's Theory of Failure, also known as the Maximum Principal Strain Theory, a material will fail (yield) when:

Options :

1. the maximum principal stress at any point in the material reaches the yield strength determined from a simple uniaxial tensile test
2. the maximum shear stress at any point in the material reaches the shear yield strength determined from a simple uniaxial tensile test
3. the total strain energy per unit volume at any point reaches the strain energy per unit volume at the yield point in a simple uniaxial tensile test
4. the maximum principal strain at any point in the material reaches the strain at the yield point determined from a simple uniaxial tensile test

Question Number : 27 Question Id : 4410091172419 Question Type : MCQ Option Shuffling : No

St. Venant's theory predicts failure based on:

Options :

1. Maximum stress
2. Maximum strain
3. Total energy
4. Shear strain

Question Number : 28 Question Id : 4410091472410 Question Type : MCQ Option Shuffling : No

As per Haigh's theory, how is the maximum strain energy per unit volume determined when it reaches its limiting value?

Options :

1. By theoretical calculation ignoring material properties
2. By a simple stress test up to the point of failure
3. By measuring only axial displacement of the member
4. By measuring elongation in a single direction

Question Number : 29 Question Id : 4410091472445 Question Type : MCQ Option Shuffling : No

According to Saint Venant's theory, a member fails or yields under which condition?

Options :

1. When the total energy of deformation surpasses the critical value
2. When the average of all principal strains reaches its maximum value
3. When hydrostatic stress in the member reaches the ultimate limit
4. When the maximum principal (or normal) strain reaches its limit

Question Number : 30 Question Id : 4410091478660 Question Type : MCQ Option Shuffling : No

Which of the following statements is true about caulking and fullering?

Options :

1. Caulking is used to finish the weld bead, whereas fullering is used only for decorative purposes.
2. Caulking and fullering are both used to make welded joints leak-proof by compressing the plate edges.
3. Caulking removes excess metal from a weld, whereas fullering increases the gap between plates.
4. Caulking is performed after painting, whereas fullering must be done before assembly.

Question Number : 31 Question Id : 4410091352693 Question Type : MCQ Option Shuffling : No

A main pipe divides into two pipes. Length and diameter of the first pipe are 20 m and 10 cm while that of the second pipe are 20 m and 5 cm respectively. Determine the relation between velocity of the first pipe (V_1) and velocity of the second pipe if friction factor of both the pipes are the same.

Options :

1. $V_1 = V_2$
2. $V_1 = 2V_2$
3. $V_1 = \sqrt{2}V_2$
4. $\sqrt{2}V_1 = V_2$

Question Number : 32 Question Id : 4410091077469 Question Type : MCQ Option Shuffling : No

The thickness for a tube of internal diameter 100 mm is subjected to an internal pressure which is $\frac{5}{8}$ of the value of max. The hoop's stress will be_____.

Options :

1. 11.25 mm
2. 31.25 mm
3. 51.25 mm
4. 71.25 mm

Question Number : 33 Question Id : 4410091688881 Question Type : MCQ Option Shuffling : No

Which similarity requirement ensures kinematic correspondence of velocities between a model and the actual system?

Options :

1. Kinematic similarity
2. Geometric similarity
3. Dynamic similarity
4. Dimensional similarity

Question Number : 34 Question Id : 4410091353192 Question Type : MCQ Option Shuffling : No

Find the power transmitted through the nozzle having outside diameter 400 mm fitted at the end of a 100 m long and 0.5 m diameter pipe. Water flows through pipe at the rate of $2 \text{ m}^3/\text{s}$ and head of water at pipe inlet is 100 m. Take friction factor as 0.04.

Options :

1. 1235.45 kW
2. 113.42 kW
3. 123.54 kW
4. 1134.26 kW

Question Number : 35 Question Id : 4410091453156 Question Type : MCQ Option Shuffling : No

For a fluid element moving in a curved pipe, which combination correctly represents the dominant forces?

Options :

1. Pressure force and gravitational force only
2. Viscous force and turbulent shear force only
3. Pressure force, viscous force, and centrifugal (inertial) force
4. Only surface tension force

Question Number : 36 Question Id : 4410091323373 Question Type : MCQ Option Shuffling : No

A pressure cooker with its steam outlet closed behaves as:

Options :

1. an open system
2. a closed system
3. an isolated system
4. a rigid system

Question Number : 37 Question Id : 4410091399188 Question Type : MCQ Option Shuffling : No

For equal maximum temperature and heat input, which cycle gives maximum efficiency?

Options :

1. Otto Cycle
2. Diesel Cycle
3. Dual Cycle
4. Brayton Cycle

Question Number : 38 Question Id : 4410091447093 Question Type : MCQ Option Shuffling : No

In a dual cycle, heat addition takes place _____.

Options :

1. At constant volume
2. At constant pressure
3. At constant volume then at constant pressure
4. At constant pressure then at constant volume

Question Number : 39 Question Id : 4410091310532 Question Type : MCQ Option Shuffling : No

In a closed system, 1 kg of gas undergoes a process in which the internal energy is reduced by 100 kJ, while the system does boundary work of 60 kJ. Another 50 kJ of boundary work done is on the system to reach the system at its initial stage. What will be the heat supplied to the system in whole scenario?

Options :

1. 110 kJ
2. 10 kJ
3. -90 kJ
4. -110 kJ

Question Number : 40 Question Id : 4410091455551 Question Type : MCQ Option Shuffling : No

A heat pump delivers 150 kJ of heat to a room and absorbs 120 kJ from the outside. What is the **Coefficient of Performance (COP)** of the heat pump?

Options :

1. 3
2. 4
3. 5
4. 6

Question Number : 41 Question Id : 441009192360 Question Type : MCQ Option Shuffling : No

Which of the following formulas is used to estimate the force required for upsetting under ideal open-die forging conditions, where deformation is homogeneous, die surfaces have no friction, and the flow stress is constant?

Where:

- F = force required
- Y_f = flow stress
- A = cross-sectional area of the part

Options :

1. $F = Y_f \cdot A$
2. $F = \frac{Y_f}{A}$
3. $F = Y_f + A$
4. $F = Y_f \cdot A^2$

Question Number : 42 Question Id : 4410091172947 Question Type : MCQ Option Shuffling : No

Which of the following types of cores is used to produce a blind hole along a horizontal axis in a casting?

Options :

1. Vertical core
2. Balanced core
3. Hanging core
4. Wire core

Question Number : 43 Question Id : 4410091313453 Question Type : MCQ Option Shuffling : No

Which extrusion process is preferred for brittle materials like tungsten?

Options :

1. Direct extrusion
2. Indirect extrusion
3. Hydrostatic extrusion
4. Impact extrusion

Question Number : 44 Question Id : 4410091169182 Question Type : MCQ Option Shuffling : No

Which of the following is an allowance given to compensate for metal shrinkage during solidification?

Options :

1. Draft allowance
2. Shrinkage allowance
3. Machining allowance
4. Distortion allowance

Question Number : 45 Question Id : 4410091375337 Question Type : MCQ Option Shuffling : No

A key advantage of cold working over hot working is:

Options :

1. no need for high-temperature furnaces
2. complete elimination of strain hardening
3. easier metal flow and lower forces
4. reduced power requirement

Question Number : 46 Question Id : 4410091375402 Question Type : MCQ Option Shuffling : No

A key disadvantage of cold working is:

Options :

1. requirement of high temperature furnaces
2. formation of surface oxidation and scaling
3. inability to achieve close dimensional accuracy
4. requirement of high forces and powerful equipment

Question Number : 47 Question Id : 4410091464221 Question Type : MCQ Option Shuffling : No

How does the equipment design differ between indirect and direct extrusion?

Options :

1. Indirect uses a simpler ram system
2. Direct requires a moving die mechanism
3. Indirect requires a moving die mechanism
4. Equipment design is identical

Question Number : 48 Question Id : 4410091684937 Question Type : MCQ Option Shuffling : No

What is the most significant mechanical property change observed in metals after a single pass through a wire drawing die?

Options :

1. Increase in tensile strength due to strain hardening.
2. Increase in ductility due to elongation.
3. Decrease in melting point due to deformation.
4. Increase in electrical conductivity due to alignment.

Question Number : 49 Question Id : 441009192389 Question Type : MCQ Option Shuffling : No

Which of the following best describes the cluster rolling process used in bulk metal forming?

Options :

1. Rolling process using only two large diameter rolls for high reduction.
2. Rolling process using small diameter rolls supported by multiple backup rolls.
3. Rolling process using tapered rolls for reducing conical shapes.
4. Rolling process using rolls that rotate in opposite directions without backup rolls.

Question Number : 50 Question Id : 441009674557 Question Type : MCQ Option Shuffling : No

Which of the following is a primary disadvantage of cold working processes?

Options :

1. Poor surface finish due to oxide scaling
2. Requires reheating between stages due to high ductility
3. Causes grain refinement during deformation
4. Limited to materials with good ductility and may cause strain hardening

Question Number : 51 Question Id : 4410091375452 Question Type : MCQ Option Shuffling : No

If the flow stress of a material doubles when the true strain increases four times, the strain hardening exponent n is approximately:

Options :

1. 0.25
2. 0.5
3. 1.0
4. 2.0

Question Number : 52 Question Id : 4410091478004 Question Type : MCQ Option Shuffling : No

For open-die forging, the maximum forging load can be estimated by:

Options :

1. $P_{\max} = kA \left(1 + \frac{2\mu h}{3r} \right)$
2. $P_{\max} = kA$
3. $P_{\max} = 2kA \left(\frac{r}{h} \right)$
4. $P_{\max} = kA(1 - \mu^2)$

Question Number : 53 Question Id : 4410091478016 Question Type : MCQ Option Shuffling : No

If a material has a flow curve defined by $\sigma = 800\varepsilon^{0.2}$, the true stress at $\varepsilon = 0.3$ will be approximately:

Options :

1. 800 MPa
2. 630 MPa
3. 560 MPa
4. 920 MPa

Question Number : 54 Question Id : 4410092095946 Question Type : MCQ Option Shuffling : No

A lap adhesive joint is redesigned by increasing overlap length while maintaining the same adhesive material and thickness. The joint strength improves because the applied load is distributed over a larger bonded region. Which principle explains this improvement?

Options :

1. Reduced average tensile stress across adhesive area
2. Reduced average thermal stress across adhesive area
3. Reduced average residual stress across adhesive area
4. Reduced average shear stress across adhesive area

Question Number : 55 Question Id : 4410092095947 Question Type : MCQ Option Shuffling : No

A rotating-disc atomization process is employed for producing alloy powder. During operation, the disc rotational speed is increased substantially while maintaining the melt feed rate. Which outcome is expected from the atomization stage?

Options :

1. Finer droplets generated due to higher centrifugal forces
2. Coarser droplets generated due to higher centrifugal forces
3. Finer droplets generated due to lower centrifugal forces
4. Coarser droplets generated due to lower centrifugal forces

Question Number : 56 Question Id : 4410091000303 Question Type : MCQ Option Shuffling : No

What is the defining characteristic of an orthogonal machining operation?

Options :

1. The cutting edge is parallel to the direction of tool travel.
2. The cutting edge is perpendicular to the direction of tool travel.
3. The tool moves perpendicular to the workpiece axis.
4. The cutting force acts parallel to the chip flow.

Question Number : 57 Question Id : 4410091284220 Question Type : MCQ Option Shuffling : No

According to Taylor's tool life equation, increasing cutting speed results in:

Options :

1. Increased tool life
2. Reduced tool life
3. No effect on tool life
4. Increased feed rate

Question Number : 58 Question Id : 4410091311395 Question Type : MCQ Option Shuffling : No

Which type of wear affects surface finish the most?

Options :

1. Crater wear
2. Flank wear
3. Notch wear
4. Nose wear

Question Number : 59 Question Id : 4410091311308 Question Type : MCQ Option Shuffling : No

Multi-point tools need a clearance angle in order to:

Options :

1. provide strength
2. avoid rubbing
3. support rake
4. reduce vibrations

Question Number : 60 Question Id : 4410091479351 Question Type : MCQ Option Shuffling : No

In orthogonal cutting, the cutting edge of the tool is—

Options :

1. Perpendicular to the cutting direction
2. Inclined slightly toward the feed motion
3. Oriented parallel to the primary cutting path
4. Tilted marginally away from the chip flow

Question Number : 61 Question Id : 4410091482645 Question Type : MCQ Option Shuffling : No

In subtractive manufacturing, the final product is obtained by:

Options :

1. Removing excess material from a workpiece
2. Adding successive layers of material onto a substrate
3. Depositing molten polymers to build complex shapes
4. Assembling prefabricated components into a structure

Question Number : 62 Question Id : 4410091482735 Question Type : MCQ Option Shuffling : No

The electrolyte commonly used in ECM for machining steel components is:

Options :

1. Sodium nitrate (NaNO_3) solution
2. Sodium hydroxide (NaOH) solution
3. Hydrochloric acid (HCl) solution
4. Sulfuric acid (H_2SO_4) solution

Question Number : 63 Question Id : 4410091311341 Question Type : MCQ Option Shuffling : No

The dominant factor affecting tool life is:

Options :

1. cutting speed
2. feed rate
3. depth of cut
4. coolant

Question Number : 64 Question Id : 4410091315363 Question Type : MCQ Option Shuffling : No

Material removal in Abrasive Jet Machining (AJM) mainly depends on:

Options :

1. workpiece ductility
2. abrasive hardness and velocity
3. tool sharpness
4. cutting fluid properties

Question Number : 65 Question Id : 4410091360107 Question Type : MCQ Option Shuffling : No

Which of the following is an advantage of Laser Beam Machining over traditional machining?

Options :

1. No heat-affected zones (HAZ) on the workpiece
2. Capability to machine hard and complex-shaped materials without tool wear
3. Requires electrical conductivity of workpiece
4. High mechanical contact forces needed

Question Number : 66 Question Id : 4410091377345 Question Type : MCQ Option Shuffling : No

In a high-speed orthogonal cutting operation, the chip thickness ratio is observed to decrease as the uncut chip thickness increases for the same tool rake angle and work material. Which explanation is most appropriate for this observation?

Options :

1. Shear angle increases with greater uncut chip thickness
2. Friction at the tool-chip interface decreases with greater uncut chip thickness
3. Shear plane area becomes larger as uncut chip thickness increases
4. Tool wear directly increases chip thickness ratio

Question Number : 67 Question Id : 4410091479279 Question Type : MCQ Option Shuffling : No

In sintered carbide tools, excessive cobalt binder content generally results in—

Options :

1. Increased hardness but reduced toughness
2. Decreased hardness but improved toughness
3. Enhanced chemical stability at high speeds
4. Greater tendency to plastic deformation under load

Question Number : 68 Question Id : 4410091482624 Question Type : MCQ Option Shuffling : No

In metal cutting operations, notch wear on a cutting tool typically appears:

Options :

1. On the flank face exactly at the depth-of-cut line
2. As a large crater formed on the rake face surface
3. Uniformly around the rounded nose of the cutting edge
4. Specifically on ceramic tools under high-speed cutting

Question Number : 69 Question Id : 4410091482681 Question Type : MCQ Option Shuffling : No

If cutting force (F_c) = 1000 N, thrust force (F_t) = 400 N, and rake angle (α) = 10° , then the frictional force (F) is approximately:

Options :

1. 568 N
2. 475 N
3. 624 N
4. 345 N

Question Number : 70 Question Id : 4410091482741 Question Type : MCQ Option Shuffling : No

The tool wear rate in electrical discharge machining (EDM) increases with:

Options :

1. Higher pulse duration and current
2. Lower current density and voltage
3. Reduced dielectric flushing pressure
4. Decrease in discharge frequency

Question Number : 71 Question Id : 441009188172 Question Type : MCQ Option Shuffling : No

The term “limits of size” in inspection refers to _____

Options :

1. Physical boundaries of a machine
2. Extreme permissible dimensions
3. Location of measurement marks
4. Standard measuring time

Question Number : 72 Question Id : 4410091480465 Question Type : MCQ Option Shuffling : No

What is the main purpose of providing tolerance in engineering components?

Options :

1. To eliminate all dimensional variations during manufacturing.
2. To achieve perfect accuracy in every produced part.
3. To allow acceptable dimensional variation for proper fit.
4. To reduce flexibility in manufacturing processes.

Question Number : 73 Question Id : 4410091495583 Question Type : MCQ Option Shuffling : No

In an internal feature like hole, GO Plug Gauge dimension will be equal to _____.

Options :

1. maximum permissible dimension of hole
2. minimum permissible dimension of hole
3. average dimension of hole
4. dimension of NO-GO gauge

Question Number : 74 Question Id : 4410091284275 Question Type : MCQ Option Shuffling : No

A hole–shaft pair is given as 30 H7/g6.

Hole: 30.000 to 30.025 mm

Shaft: 29.959 to 29.972 mm

What is the maximum clearance at MMC condition?

Options :

1. 0.028 mm
2. 0.066 mm
3. 0.050 mm
4. 0.059 mm

Question Number : 75 Question Id : 4410091495606 Question Type : MCQ Option Shuffling : No

Which among the following is the primary purpose of using No-Go gauges in an inspection process?

Options :

1. To ensure part is within minimum material limit
2. To ensure part is within maximum material limit
3. To get precise dimensional measurements
4. To measure the surface roughness of the component

Question Number : 76 Question Id : 4410091495774 Question Type : MCQ Option Shuffling : No

In metrology, tolerance is a measure of the _____.

Options :

1. accuracy of measuring instrument
2. permissible variation in dimension
3. cost of manufacturing
4. expected wear of a part

Question Number : 77 Question Id : 4410092095949 Question Type : MCQ Option Shuffling : No

A company is required to replace a product for a customer due to a malfunction discovered after delivery. Which category of Cost of Quality does this replacement illustrate?

Options :

1. External failure cost
2. Internal failure cost
3. Prevention cost
4. Appraisal cost

Question Number : 78 Question Id : 4410092095953 Question Type : MCQ Option Shuffling : No

A firm's acceptance sampling plan occasionally rejects good lots and accepts bad lots. Which two types of errors are associated with these outcomes?

Options :

1. Confidence interval and acceptance range
2. Sampling bias and inspection error
3. Defective rate and pass rate
4. Producer's risk and consumer's risk

Question Number : 79 Question Id : 4410092095950 Question Type : MCQ Option Shuffling : No

In a high-volume manufacturing plant, a team is asked to identify which statistical quality control chart should be used to monitor the proportion of defective items produced daily. Which of the following control charts is appropriate for this purpose?

Options :

1. C-chart
2. X-bar chart
3. R-chart
4. P-chart

Question Number : 80 Question Id : 4410092095959 Question Type : MCQ Option Shuffling : No

A production system records the following annual data: planned preventive maintenance downtime = 120 hours, breakdown downtime = 40 hours. After revising the preventive maintenance schedule, planned downtime increases to 150 hours while breakdown downtime decreases to 5 hours. What is the primary justification for this change?

Options :

1. Increased downtime provides greater inventory flexibility than forecasting
2. Planned downtime provides greater process capability than inspections
3. Scheduled downtime provides greater utilization than productivity improvement
4. Controlled downtime provides greater reliability than unexpected failures

Question Number : 81 Question Id : 4410092095976 Question Type : MCQ Option Shuffling : No

In a Standard Data System, basic motion elements are used to determine the time required for a manual task. Which feature of the system helps achieve this?

Options :

1. Grouping of common work elements
2. Performing visual inspection only
3. Using random time sampling
4. Customizing each task separately

Question Number : 82 Question Id : 4410092095978 Question Type : MCQ Option Shuffling : No

Which type of allowance compensates for the physical or mental tiredness that accumulates during repetitive tasks in a time study?

Options :

1. Fatigue allowance
2. Personal allowance
3. Contingency allowance
4. Special allowance

Question Number : 83 Question Id : 4410092095977 Question Type : MCQ Option Shuffling : No

A production manager chooses a Predetermined motion time system (PMTS) to design a new workstation layout. What benefit does this offer compared to work sampling techniques?

Options :

1. It enables prediction of task times for hypothetical layouts
2. It is based on previously observed worker performance
3. It requires fewer motion breakdowns than work sampling
4. It relies on actual output rates from existing layouts

Question Number : 84 Question Id : 4410092095989 Question Type : MCQ Option Shuffling : No

In a facility where fragile electronic components need to be transferred between assembly lines without vibration, which material handling equipment is best suited to minimize potential damage?

Options :

1. Belt conveyor
2. Chain conveyor
3. Roller conveyor
4. Pallet jack

Question Number : 85 Question Id : 4410092095990 Question Type : MCQ Option Shuffling : No

A manufacturing facility needs to move small uniform boxes between two fixed points over a long distance with minimal manual intervention. Which conveyor system would be the most appropriate to select for this purpose?

Options :

1. Chain conveyor system
2. Belt conveyor system
3. Roller conveyor system
4. Screw conveyor system

Question Number : 86 Question Id : 4410092095962 Question Type : MCQ Option Shuffling : No

A manufacturing organization evaluates two alternative designs. Design P has a lower manufacturing cost but significantly higher warranty expenses after product release. Design Q has a slightly higher manufacturing cost but substantially lower warranty expenses. Which design should be selected according to sound cost consideration principles?

Options :

1. Design P because manufacturing expenditure remains comparatively lower overall
2. Design Q because total lifecycle expenditure becomes comparatively lower
3. Design P because warranty expenditure occurs after production completion
4. Design Q because manufacturing expenditure exceeds procurement expenditure

Question Number : 87 Question Id : 4410092095964 Question Type : MCQ Option Shuffling : No

A gearbox redesign project shows that a high-cost hardened steel gear and a lower-cost alloy gear both satisfy torque transmission requirements, but differ in maintenance frequency and lifecycle replacement cost. The decision team evaluates value based on functional performance stability over time. Which decision logic is most appropriate?

Options :

1. Selection based on lifecycle cost per functional unit of torque transmission over operational duration
2. Selection based on lowest initial manufacturing cost regardless of maintenance implications
3. Selection based on maximum hardness material irrespective of operational environment variation
4. Selection based on minimum machining time required for gear production cycle efficiency

Question Number : 88 Question Id : 4410092095966 Question Type : MCQ Option Shuffling : No

Two product designs satisfy identical functional requirements. Design P requires five-axis machining from solid stock, whereas Design Q requires casting followed by limited finish machining. Annual production volume is expected to exceed 300,000 units. Which alternative is better aligned with design for manufacturing (DFM) principles?

Options :

1. Design P because dimensional flexibility increases significantly at higher production volumes
2. Design Q because machining accuracy increases significantly at higher production volumes
3. Design P because tooling investment decreases significantly at higher production volumes
4. Design Q because processing cost decreases significantly at higher production volumes

Question Number : 89 Question Id : 4410092095971 Question Type : MCQ Option Shuffling : No

In a scenario where process mapping reveals excessive backtracking of materials, which method study principle can be most effectively applied to resolve this inefficiency?

Options :

1. Combination of related operations into a single process
2. Standardization of work methods across operators
3. Mechanization of material handling activities
4. Rearrangement of work sequence or layout

Question Number : 90 Question Id : 4410092095975 Question Type : MCQ Option Shuffling : No

A firm introduces work sampling to evaluate different process activities. Which advantage does this technique offer compared to continuous time study in a complex environment?

Options :

1. It provides more accurate cycle times for repetitive operations
2. It eliminates the need for performance rating during analysis
3. It determines the exact time required for every work cycle
4. It allows data collection without constant observer presence

Question Number : 91 Question Id : 4410091228967 Question Type : MCQ Option Shuffling : No

What is Slack or Float in CPM (critical path method)?

Options :

1. The amount of time an activity can be delayed without affecting the project duration
2. The amount of time an activity must be completed before the next activity starts
3. The amount of resources allocated to an activity
4. The amount of dependencies between activities

Question Number : 92 Question Id : 4410091260058 Question Type : MCQ Option Shuffling : No

Which of the following options is correct about exponential smoothing method of forecasting?

Options :

1. Gives equal weight to all past data
2. Gives more weight to recent data
3. Ignores past data completely
4. Only works with moving averages

Question Number : 93 Question Id : 441009336982 Question Type : MCQ Option Shuffling : No

Exponential smoothing with trend is also called _____.

Options :

1. simple linear regression
2. double exponential smoothing
3. triple exponential smoothing
4. logarithmic smoothing

Question Number : 94 Question Id : 4410091049180 Question Type : MCQ Option Shuffling : No

Consider a transportation problem with three sources (rows) and four demand points. What will be the total cost of using the Minimum Cost Method?

4	6	8	8	40
6	8	6	7	60
5	7	6	8	50
20	30	50	50	

Options :

1. 960
2. 980
3. 970
4. 950

Question Number : 95 Question Id : 4410091386990 Question Type : MCQ Option Shuffling : No

In Weighted Moving Average, increasing weight for recent data generally:

Options :

1. Makes forecast more responsive to changes
2. Makes forecast less sensitive
3. Makes forecast constant over time
4. Increases lag in forecast

Question Number : 96 Question Id : 4410091066993 Question Type : MCQ Option Shuffling : No

Which statement is true regarding the Simplex Method's computational procedure?

Options :

1. It starts at an initial basic feasible solution and moves to adjacent corner points.
2. It guarantees the global maximum for non-linear problems.
3. It begins at a random point and explores outward.
4. It tests all possible solutions in the feasible region simultaneously.

Question Number : 97 Question Id : 4410091259945 Question Type : MCQ Option Shuffling : No

In a PERT network, the time estimates for an activity are: Optimistic time (a) = 4 days, Most likely time (m) = 10 days & Pessimistic time (b) = 16 days. What is the standard deviation of the activity duration?

Options :

1. 6
2. 4
3. 2
4. 0

Question Number : 98 Question Id : 4410091386206 Question Type : MCQ Option Shuffling : No

The Critical Path Method primarily assumes which type of time estimates for activities?

Options :

1. Single deterministic time estimate
2. Three-point probabilistic estimates
3. Variable cost-based estimates
4. Randomly distributed time estimates

Question Number : 99 Question Id : 4410091386218 Question Type : MCQ Option Shuffling : No

Given activities with optimistic time 4 days, most likely time 6 days, and pessimistic time 10 days, what is the expected activity time in PERT?

Options :

1. 6.33 days
2. 6 days
3. 7 days
4. 8 days

Question Number : 100 Question Id : 4410091386961 Question Type : MCQ Option Shuffling : No

The smoothing constant α in exponential smoothing represents:

Options :

1. Weight given to latest actual observation
2. Average error magnitude
3. Rate of data accumulation
4. Variance of forecast errors